Arsenic in seaweed--forms, concentration and dietary exposure.

This study has measured the content of total and inorganic forms of arsenic in seaweed available on retail sale for consumption, to provide data for dietary exposure estimates and to support advice to consumers. A total of 31 samples covering five varieties of seaweed were collected from various retail outlets across London and the internet. All of the samples were purchased as dried product. For four of the five varieties, soaking was advised prior to consumption. The recommended method of preparation for each individual sample was followed, and total and inorganic arsenic were analysed both before and after preparation. The arsenic remaining in the water used for soaking was also measured. Arsenic was detected in all samples with total arsenic at concentrations ranging from 18 to 124 mg/kg. Inorganic arsenic, which can cause liver cancer, was only found in the nine samples of hijiki seaweed that were analysed, at concentrations in the range 67-96 mg/kg. Other types of seaweed were all found to contain less than 0.3mg/kg inorganic arsenic, which was the limit of detection for the method used. Since consumption of hijiki seaweed could significantly increase dietary exposure to inorganic arsenic, the UK Food Standards Agency (FSA) issued advice to consumers to avoid eating it.